MATH 440/540 **HW 12** Schreyer

Due Thursday Nov. 7

WITH SOLUTIONS

Please follow the homework instructions listed in HW 1 for writing up your homework.

1. (12 pts) Z& T Chapter 7, Section 16:

   (a) (4 pts) Carefully write a "well-posed" Neumann problem for the Laplace equation. Why is "well-posed" in quotes?

   Let Ω be a regular (bounded, Lipschitz boundary, domain). Find $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ such that

$$\begin{aligned} &\triangle u = 0, && \Omega \\ &\frac{\partial u}{\partial n} = f(\boldsymbol{x}), && \partial\Omega \end{aligned}$$

   where $f(x) \in C^0(\partial\Omega)$ and $\int_{\partial\Omega} f(\boldsymbol{x}) d\sigma = 0$.

   Why is "well-posed" in quotes? Because uniqueness is only up to a constant.

   (b) (4 pts) Let Ω be the unit square, $0 < x < 1$, $0 < y < 1$. Give 2 examples of Neumann boundary conditions for which we *are not* guaranteed at least one solution. Explain why we are not guaranteed at least one solution.

   Note: If $f(x)$ is not continuous, we could have no solution, but we could also have many solutions. So the safe bet is to construct two functions, $f(x)$, so that the integral of the function about the boundary is not zero. This integral should be done *counterclockwise*, since the restriction comes from using the divergence theorem. The function can be discontinuous or continuous. Here are two examples:

$$f(x,y) = \begin{cases} 1, & 0 \le x < 1,\ y = 0 \\ 2, & x = 1,\ 0 \le y < 1 \\ 3, & 0 \le x < 1,\ y = 1 \\ 4, & x = 0,\ 0 \le y < 1 \end{cases}$$

$$\int_{\partial\Omega} f(\boldsymbol{x}) d\sigma = \int_{x=0,y=0}^{x=1} 1\, dx + \int_{y=0,x=1}^{y=1} 2\, dy + \int_{x=1,y=1}^{x=0} 3\,(-dx) + \int_{y=1,x=0}^{y=0} 4\,(-dy)$$
$$= 1 + 2 + 3 + 4 = 10$$

   which means we have a net flow out of 10 units of heat, so the system can never reach a steady-state temperature distribution.

$$f(x,y) = \begin{cases} 1, & 0 \le x < 1,\ y = 0 \\ y, & x = 1,\ 0 \le y < 1 \\ 1, & 0 \le x < 1,\ y = 1 \\ y^2, & x = 0,\ 0 \le y < 1 \end{cases}$$

$$\int_{\partial\Omega} f(\boldsymbol{x}) d\sigma = \int_{x=0,y=0}^{x=1} 1\ dx + \int_{y=0,x=1}^{y=1} y\ dy + \int_{x=1,y=1}^{x=0} 1\ (-dx) + \int_{y=1,x=0}^{y=0} y^2\ (-dy)$$

$$= 1 + \frac{1}{2} + 1 + \frac{1}{3} = 2\frac{5}{6}$$

which means we have a net flow of 2 5/6 units of heat, so the system can never reach a steady-state temperature distribution.

(c) (4 pts) Let Ω be the unit square, $0 < x < 1$, $0 < y < 1$. Give 2 examples of Neumann boundary conditions for which we *are* guaranteed at least one solution. Explain.

Strictly, $f(x)$ should be continuous. One could come up with a harmonic function and calculate $\boldsymbol{\nabla} u \cdot \boldsymbol{n}$ on the boundaries, or guess and check. Here are two examples, the second example coming from the harmonic function $u = xy$.

$$f(x,y) = \begin{cases} 0, & 0 \le x < 1,\ y = 0 \\ 0, & x = 1,\ 0 \le y < 1 \\ 0, & 0 \le x < 1,\ y = 1 \\ 0, & x = 0,\ 0 \le y < 1 \end{cases}$$

$$\int_{\partial\Omega} f(\boldsymbol{x}) d\sigma = 0$$

Consider $u = xy$, which is harmonic. Then $\boldsymbol{\nabla} u =< y,\ x >$. So on the bottom boundary, $y = 0$ and $\boldsymbol{n} =< -1,\ 0 >$ so $f(x) = \boldsymbol{\nabla} u \cdot \boldsymbol{n} =< 0, x > \cdot < 0, -1 >= -x$ on the bottom boundary. Continue to get

$$f(x,y) = \begin{cases} -x, & 0 \le x < 1,\ y = 0 \\ y, & x = 1,\ 0 \le y < 1 \\ x, & 0 \le x < 1,\ y = 1 \\ -y, & x = 0,\ 0 \le y < 1 \end{cases}$$

$$\int_{\partial\Omega} f(\boldsymbol{x}) d\sigma = \int_{x=0,y=0}^{x=1} -x\ dx + \int_{y=0,x=1}^{y=1} y\ dy + \int_{x=1,y=1}^{x=0} x\ (-dx) + \int_{y=1,x=0}^{y=0} -y\ (-dy)$$

$$= -\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2} = 0$$

2. (8 pts) Z&T Chapter 7, 16.1 Find *all* solutions to the following Neumann problem on the unit circle:

$$\triangle u = 0, \qquad 0 \le r < 1, \quad -\pi \le \theta \le \pi$$
$$\frac{\partial u}{\partial n}(1,\theta) = f(\theta) = |\theta| - \frac{\pi}{2}, \quad -\pi \le \theta \le \pi$$

Be sure to check that a solution does exist (!)

First check that a solution does exist: $f \in C^0(\partial\Omega)$, yes. $\int_{-\pi}^{\pi} f(\theta)(1)d\theta = 0$, yes. Next find the Fourier series of $f(\theta)$. Since $f(\theta)$ is even, we know the coefficients of the sine terms, $b_n = 0$. The first coefficient, $a_0/2$ is the average value of $f(\theta)$, and this is zero (same calculation as for whether $f(\theta)$ satisfies the integral requirement). So we are left with a_n. Using the fact that an even function times an even function is an even function:

$$\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi}\left[|\theta| - \frac{\pi}{2}\right]\cos(n\theta)\,d\theta \\
&= \frac{2}{\pi}\int_{0}^{\pi}\left[|\theta| - \frac{\pi}{2}\right]\cos(n\theta)\,d\theta \\
&= \frac{2}{\pi}\int_{0}^{\pi}\left[\theta - \frac{\pi}{2}\right]\cos(n\theta)\,d\theta \\
&= \frac{2}{\pi}\left[\left(\theta - \frac{\pi}{2}\right)\left(\frac{1}{n}\sin(n\theta)\right)\Big|_{\theta=0}^{\pi} - \int_{\theta=0}^{\pi}\frac{1}{n}\sin(n\theta)\,d\theta\right] \\
&= \frac{2}{\pi}\left[0 + \frac{1}{n^2}\cos(n\theta)\Big|_{\theta=0}^{\pi}\right] \\
&= \frac{2}{\pi n^2}\left[(-1)^n - 1\right]
\end{aligned}$$

In order to satisfy the boundary condition we need the coefficients of the solution, A_n, to be equal to $\frac{1}{n}a_n$ so that (keeping in mind that *the solution is unique only up to a constant*, so we bring in A_0):

$$\begin{aligned}
u(r,\theta) &= \frac{A_0}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{[(-1)^n - 1]}{n^3}r^n\cos(n\theta) \\
&= \frac{A_0}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{r^{2k-1}}{(2k-1)^3}\cos[(2k-1)\theta]
\end{aligned}$$

3. (4 pts) Z&T Chapter 8, 1.1 Show that $u(x_1, ..., x_n, t) = F(\xi_1 x_1 + ... + \xi_n x_n - t)$ where $\boldsymbol{\xi} = (\xi_1, ..., \xi_n, -1)$ is a characteristic vector, is a solution to the wave equation: $u_{x_1x_1} + ... + u_{x_nx_n} - u_{tt} = 0$.

   First recall that $\boldsymbol{\xi}$ satisfies the characteristic equation (see Chapter 5, Section 1):

$$\xi_1^2 + \xi_2^2 + ... + \xi_n^2 - \xi_t^2 = 0.$$

   To fix a direction (because we only care about direction and not magnitude), it is custom to set $\xi_t = 1$.

We next calculate the partial derivatives using $u(x_1, ..., x_n, t) = F(\xi_1 x_1 + ... + \xi_n x_n - t)$:

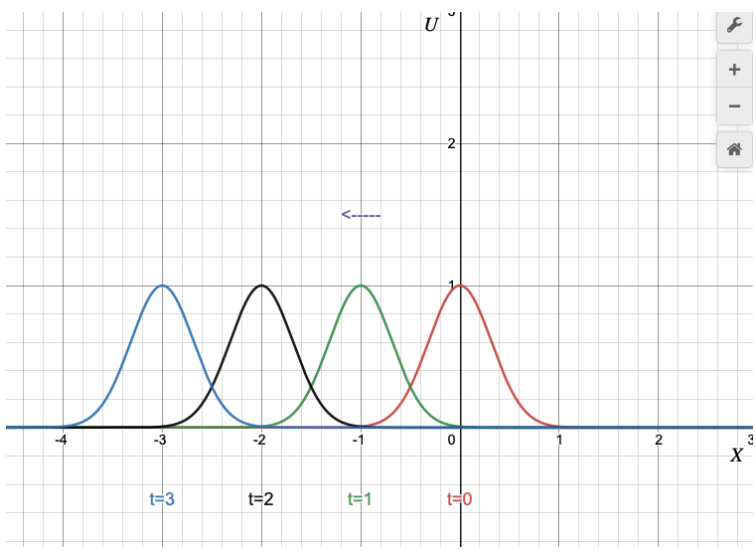
$$u_{x_1} = F'(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_1, \quad u_{x_2} = F'(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_2, \ ...$$
$$u_{x_n} = F'(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_n, \quad u_t = F'(\xi_1 x_1 + ... + \xi_n x_n - t)(-1)$$
$$u_{x_1 x_1} = F''(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_1^2, \quad u_{x_2 x_2} = F''(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_2^2, \ ...$$
$$u_{x_n x_n} = F''(\xi_1 x_1 + ... + \xi_n x_n - t)\xi_n^2, \quad u_{tt} = F''(\xi_1 x_1 + ... + \xi_n x_n - t)(-1)^2$$

Substituting into the wave equation gives:

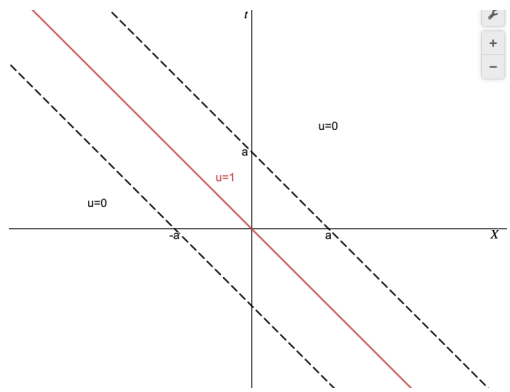
$$\begin{aligned} u_{x_1 x_1} + u_{x_2 x_2} + ... + u_{x_n x_n} - u_{tt} &= F''(\xi_1 x_1 + ... + \xi_n x_n - t)\left(\xi_1^2 + \xi_2^2 + ... + \xi_n^2 - 1\right) \\ &= F''(\xi_1 x_1 + ... + \xi_n x_n - t)(0) = 0. \end{aligned}$$

4. (8 pts) Z&T Chapter 8, 1.2 Draw solutions to the 1-dimensional wave equation of the form $u(x,t) = G(x+t)$ where $G(y) = F(y)$ is the "blip" function ($F(y)$ is 1 when $y = 0$ and goes to zero so that $F(y) = 0$ for $y \leq -a$ and $y \geq a$). Draw the solution at $t = 0, 1, 2$, and 3. Then sketch the region in the xt-plane where the solution is *not* zero.

We first draw the solutions for times $t = 0$, 1, 2, and 3. Using Desmos, we have:



Next we draw the regions in xt-space where the solution is not zero. We do this by considering where the argument of F is $-a$ and a, and knowing that between these lines, the function is not zero. The solid red line is where $u = 1$, and outside of the dashed lines $u = 0$. Using Desmos:

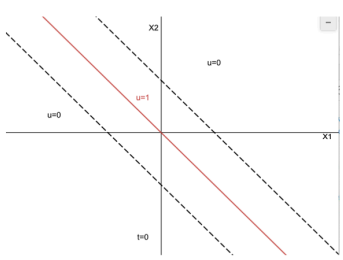


5. (6 pts) Z&T Chapter 8, 1.3 Consider the "blip" function, $F(y)$, which is 1 when $y = 0$ and goes to zero so that $F(y) = 0$ for $y \leq -a$ and $y \geq a$. Consider the solution to the 2-dimensional plane wave solution to wave equation:

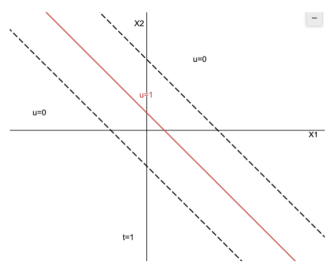
$$u(x_1,\ x_2,\ t) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - t\right).$$

In the x_1x_2 plane draw the lines where the solution has its maximum value and the strips where $u(x_1, x_2, t) \neq 0$ for $t = 0,\ 1,\ 2,\ 5$.

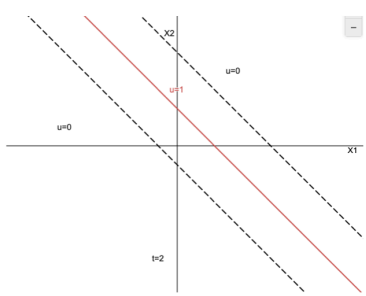
The sketches, made using Desmos, are below for times 0, 1, 2, and 5. The red line is where $u = 1$ and outside of the dashed lines $u = 0$. The distance between the two dashed lines (measured along a line orthogonal to both lines) is $2a$.



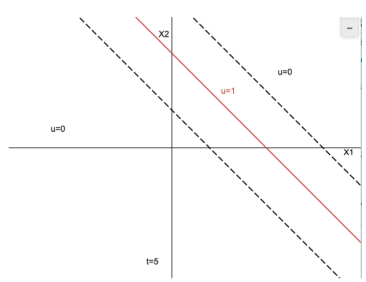
t=0



t=1



t=2



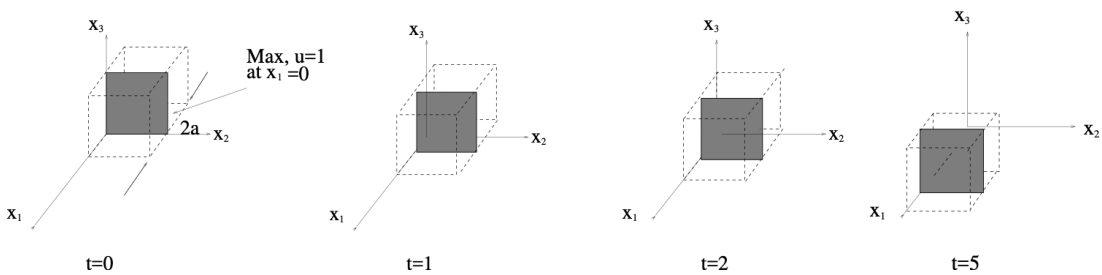
t=5

6. (6 pts) Z&T Chapter 8, 1.4 Consider the "blip" function, $F(y)$, which is 1 when $y = 0$ and goes to zero so that $F(y) = 0$ for $y \leq -a$ and $y \geq a$. Consider the the 3-dimensional plane wave solution to the wave equation:

$$u(x_1,\ x_2,\ x_3,\ t) = F\left(x_1 - t\right).$$

In $I\!R^3$ space, draw the planes where the solution has its maximum value and the slabs where $u(x_1, x_2, x_3,\ t) \neq 0$ for $t = 0,\ 1,\ 2,\ 5$.

Graphs made using XFig.



7. (6 pts) Z&T Chapter 8, 1.5 Let $< \xi_1, \xi_2, ...\xi_n, -1 >$ be a characteristic for the wave equation. We would like to show that

   (a) (3 pts) in $I\!R^{n+1}$ the planes,

   $$\xi_1 x_1 + ... + \xi_n x_n - t = c$$

   are characteristic surfaces with respect to the wave equation:

   $$\frac{\partial^2 u}{\partial x_1^2} + ... + \frac{\partial^2 u}{\partial x_n^2} - \frac{\partial^2 u}{\partial t^2} = 0.$$

   (b) (3 pts) Show that the plane wave solutions

   $$u(x_1, x_2, ..., x_n, t) = F(\xi_1 x_1 + \xi_2 x_2 + ... + \xi_n x_n - t)$$

   are constant on these planes.

(a) In $I\!R^{n+1}$ the planes,

$$\xi_1 x_1 + ... + \xi_n x_n - t = c$$

are characteristic surfaces, where $< \xi_1,\ \xi_2,\ ..., \xi_n, -1 >$ is a characteristic vector of the wave equation. To do this we need to show that the normal to the surface is a characteristic vector. Note that all variables are only to the power of 1 (no squared terms etc), so this is a plane! Recall that for an equation of a plane such as: $n_1 x + n_2 y + n_3 z = d$, the vector normal to the plane is $< n_1,\ n_2,\ n_3 >$. Thus the normal to the above surface is $< \xi_1, \xi_2, ..., \xi_n, -1 >$. Since $\boldsymbol{\xi}$ satisfies the characteristic equation for the wave equation: $\xi_1^2 + \xi_2^2 + ... + \xi_n^2 - 1 = 0$, and this vector is normal to the plane, we have shown that it is a characteristic surface.

(b) If $(x_1, x_2, ..x_n, t)$ is in the plane, then the space-time point satisifies $\xi_1 x_1 + ... + \xi_n x_n - t = c$. Thus the argument of F is a constant, i.e. for any point on the plane we have

$$u(x_1, u_2, ..., u_n, t) = F(\xi_1 x_2 + \xi_2 x_2 + ... + \xi_n x_n - t) = F(c) = C_2$$

where C_2 is a constant. Thus all points on the plane give a value of u that is C_2.

8. Z&T (4 pts) Chapter 8, 1.6 Rewrite the spherical wave equation (why is this called the *spherical* wave equation?):

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) - \frac{\partial^2 u}{\partial t^2} = 0, \quad (n = 3)$$

as a one-dimensional wave equation: $w_{rr} - w_{tt} = 0$ using the change of *dependent* variables: $w = ru$.

The idea is to replace all occurrences of u with w. So in particular, $u = \frac{1}{r}w$. Then:

$$\begin{aligned}
\frac{\partial u}{\partial r} &= -\frac{1}{r^2}w + \frac{1}{r}\frac{\partial w}{\partial r} \\
r^2\frac{\partial u}{\partial r} &= -w + r\frac{\partial w}{\partial r} \\
\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) &= -\frac{\partial w}{\partial r} + \frac{\partial w}{\partial r} + r\frac{\partial^2 w}{\partial r^2} = r\frac{\partial^2 w}{\partial r^2} \\
\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) - \frac{\partial^2 u}{\partial t^2} &= \frac{1}{r}\frac{\partial^2 w}{\partial r^2} - \frac{1}{r}\frac{\partial^2 w}{\partial t^2}.
\end{aligned}$$

If $r = 0$ (the origin) is not in the domain, then we can multiply through by r and we have the desired result.

9. (6 pts) Z& T Chapter 8, 2.1 Consider the initial value problem (IVP) for the one-dimensional wave equation:

$$\begin{aligned}
&u_{xx} - u_{tt} = 0, \qquad x \in I\!R, \quad t > 0 \\
&u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x), \qquad x \in I\!R
\end{aligned}$$

The solution is given by the d'Alembert's solution:

$$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)\, d\tau$$

(a) (3 pts) Suppose that $\phi = \psi = 0$ in some open interval of length ℓ. Up to what time t can you be sure that $u = 0$ at the center of the interval? Hint: Since you can determine where the coordinate system is, make your life easier and put $x = 0$ at the center of the interval, so that the interval where $\phi = \psi = 0$ is $-\ell/2 < x < \ell/2$.

(b) (3 pts) Suppose that $\phi = \psi = 0$ *outside* of the interval $|x| \leq 1$. Up to what time can you be sure that $u = 0$ at (i) $x = 3$, (ii) $x = 10$, and (iii) $x = -5$?

(a) We want to look at the solution when $x = 0$:

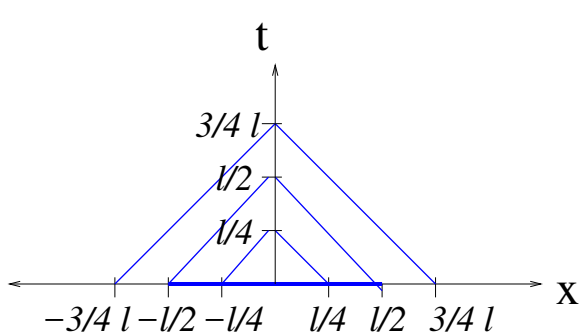
$$u(0,t) = \frac{1}{2}\left[\phi(t) + \phi(-t)\right] + \frac{1}{2}\int_{-t}^{t} \psi(\tau)\, d\tau$$

Consider the figure below. The interval on the x axis in blue, $-\ell/2 < x < \ell/2$ is where both ϕ and ψ are zero. Let's look at $t = \frac{\ell}{4}$. At that time, for the solution, we evaluate ϕ at $-\frac{\ell}{4}$ and $\frac{\ell}{4}$, but ϕ is zero there. We also evaluate ψ for all values, $-\frac{\ell}{4} \leq x \leq \frac{\ell}{4}$. But throughout the interval, ψ is zero. So we know that $u = 0$ at $x = 0$ and at time $t = \frac{\ell}{4}$.

Now consider $t = \frac{\ell}{2}$. At that time, for the solution, we evaluate ϕ at $-\frac{\ell}{2}$ and $\frac{\ell}{2}$. ϕ may not be zero there. So we are not guaranteed that this part of our solution will be zero. We also evaluate ψ for all values, $-\frac{\ell}{2} \leq x \leq \frac{\ell}{2}$. Although ψ may be not zero at the end points, since the value of an integral does not change if the function is changed at one (or two) points, the value of the integral is still zero. Thus we may not get zero due to the contribution of ϕ in the solution.

Now consider $t = \ell$. At that time, we evaluate ϕ at $-\ell$ and ℓ. ϕ may not be zero there. So we are not guaranteed that this part of our solution will be zero. We also evaluate ψ for all values, $-\ell \leq x \leq \ell$. Since ψ may be not zero for any value of $|x| \geq \frac{\ell}{2}$ the value of the integral may not be zero. Thus we may not get zero due to the contribution of both ϕ and ψ in the solution.

Generalizing this argument, we get that we are not guaranteed $u = 0$ for time $t \geq \frac{\ell}{2}$.



(b) We consider the solution to the one-dimensional wave equation where we know that initially the solution is zero over the intervals $x < -1$ and $x > 1$ (colored blue in the figure below). The question we wish to answer is up to what time t, can we be sure that $u = 0$ at the following locations? We solve geometrically, using the figure below.

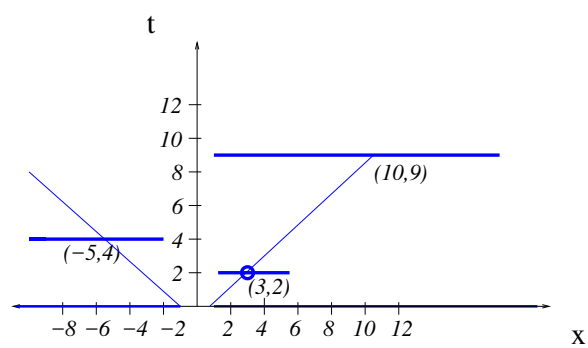
$x = 3$: The solutions is

$$u(3,t) = \frac{1}{2}\left[\phi(3+t) + \phi(3-t)\right] + \frac{1}{2}\int_{3-t}^{3+t} \psi(\tau)\, d\tau$$

At $t = 1$ the argument of $\phi(3+t)$ is 4 and the argument of $\phi(3-t)$ is 2, and the value of ϕ at both places is 0. Same thing with integrating ψ between 2 and 4 - the value of ψ is zero for all values between 2 and 4. Try $t = 2$. We get that $3 - t = 1$ and we are just entering into the domain where ϕ and ψ are not zero. For $t > 2$ we will be integrating ψ over some part or all of the interval $-1 \le x \le 1$. So we see that u is not guaranteed to be zero after $t = 2$.

$x = 10$: Since 10 is 9 units from 1, we see that it will take 9 units of time before u is not guaranteed to be zero. So u is not guaranteed to be zero after $t = 9$.

$x = -5$: Similarly, since -5 is 4 units from -1, we see that u is not guaranteed to be zero after $t = 4$.



10. (6 pts) Z& T Chapter 8, 2.2 (Hint: Change of variables.) (3 pts) Write down the formula for the solution of the initial value problem

$$\begin{aligned} &u_{xx} - \frac{1}{c^2}u_{tt} = 0, \qquad x \in I\!R, \quad t > 0 \\ &u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x), \qquad x \in I\!R \end{aligned} \tag{1}$$

Did you change the variables for the u_t initial condition too? Then answer the following questions:

(a) (3 pts) Suppose that $\phi = \psi = 0$ in some open interval of length ℓ. Up to what time t can you be sure that $u = 0$ at the center of the interval? Hint: You may assume $x = 0$ at the center of the interval (why?).

(b) (3 pts) Suppose that $\phi = \psi = 0$ *outside* of the interval $|x| \leq 1$. Up to what time can you be sure that $u = 0$ at (i) $x = 3$, (ii) $x = 10$, and (iii) $x = -5$?

We are first asked to write down the formula for the above IVP. We will use the change of variables $t' = ct$ which we obtain by guess and check so that $\frac{1}{c^2}u_{tt} = u_{t't'}$. The check on the proposed change of variables is:

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial t'}\frac{\partial t'}{\partial t} = cu_{t'}$$
$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial u_t}{\partial t}\right) = \frac{\partial}{\partial t'}\left(cu_{t'}\right)\frac{\partial t'}{\partial t} = c^2 u_{t't'}$$

Thus our IVP in terms of t' is:

$$u_{xx} - u_{t't'} = 0, \qquad x \in I\!R, \quad t' > 0$$
$$u(x,0) = \phi(x) \qquad u_t(x,0) = cu_{t'} = \psi(x), \qquad x \in I\!R$$

We know the solution to the 1-dimensional wave equation (with $c = 1$) in terms of (x, t') with initial velocity $\frac{1}{c}\psi(x)$ (have a new initial velocity from change of variables!), is the d'Alembert solution:

$$u(x,t') = \frac{1}{2}\left[\phi(x+t') + \phi(x-t')\right] + \frac{1}{2}\int_{x-t'}^{x+t'} \frac{1}{c}\psi(\tau)\, d\tau$$

Switching back to t gives:

$$u(x,t) = \frac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(\tau)\, d\tau$$

(a) We know that initially the solution is zero over the interval $[-\ell/2, \ell/2]$ where ℓ is a positive real number. We want to know up to what time t can we be sure that $u = 0$ at $x = 0$. We can solve this problem two ways:

- Algebraically: $\phi \neq 0$ when the argument of ϕ is less than (or $=$) to $-\ell/2$ or greater than (or $=$) to $\ell/2$, giving $x + ct \leq -\ell/2$ or $x + ct \geq \ell/2$.
  Let $x = 0$. Then u may not be 0 when $ct \leq \frac{-\ell}{2}$ or $ct \geq \frac{\ell}{2}$. For $t > 0$, we use the second inequality to get $t \geq \frac{\ell}{2c}$. So $u(0,t) = 0$ for $t < \frac{\ell}{2c}$.
- Change of variables using solution of previous problem. From the previous problem we have that $u = 0$ for $t' < \frac{l}{2}$. Using the change of variables: $t' = ct$ and solving for t we get $t < \frac{\ell}{2c}$.

(b) We consider the solution to (1) where we know that initially the displacement, $\phi(x)$, and velocity, $\psi(x)$, are zero over the intervals $x < -1$ and $x > 1$. The question we wish to answer is at what time is u not necessarily equal to 0 at the following locations? We solve the problem using the solution to the previous problem and performing a change of variables.

$x = 3$: From the previous problem, we have that u is not guaranteed to be zero after $t' = 2$, or $t = \dfrac{2}{c}$.

$x = 10$: From the previous problem, we have that u is not guaranteed to be zero after $t' = 9$, or $t = \frac{9}{c}$.

$x = -5$: From the previous problem, we have that u is not guaranteed to be zero after $t' = 4$, or $t = \frac{4}{c}$